

A DISCONNECTED GRACEFUL BIPARTITE GRAPH WITH NO NEAR α -VALUATION

ABSTRACT. Angus, Huczynska and McCartney ask, in the first Further Work item of arXiv:2603.05662: does every bipartite graph that possesses a β -valuation have a near α -valuation? We answer no, with a *disconnected* graph. The graph $G = C_4 \cup P_3$, on 7 vertices and 6 edges, is bipartite and graceful, and none of its 32 β -valuations is a near α -valuation; the proof is a table of 15 cases over the labels $\{0, 1, \dots, 6\}$ and uses no computer. Up to isomorphism G is the only counterexample with at most 6 edges, and 6 is the least possible number of edges. Since every counterexample found here is disconnected, the question for connected bipartite graphs is not settled by this note: each of the 1052 connected bipartite graceful isomorphism classes with at most 10 edges does have a near α -valuation.

1. THE QUESTION

Let G be a finite simple graph with m edges, no loops and no isolated vertices. Following Angus, Huczynska and McCartney [1], a β -valuation of G is an injective map

$$b: V(G) \longrightarrow \{0, 1, \dots, m\} \quad \text{such that} \\ \{ |b(u) - b(v)| : uv \in E(G) \} = \{1, 2, \dots, m\},$$

also called a *graceful labelling*; a graph admitting one is *graceful*. A β -valuation b is a *near α -valuation* if $V(G)$ admits a partition into V_{small} and V_{large} such that every vertex of V_{small} carries a label smaller than all of its neighbours' and every vertex of V_{large} a label larger than all of its neighbours'; equivalently, in the authors' own gloss, every vertex is a source or a sink. It is an α -valuation if some integer x satisfies $\min\{b(u), b(v)\} \leq x < \max\{b(u), b(v)\}$ for every edge uv . Only injectivity is required of b , so $|V(G)| \leq m + 1$; a graph with $|V(G)| = m + 1$ has each of its β -valuations a bijection onto $\{0, 1, \dots, m\}$.

The predicate is not new: it is the near α -labeling of El-Zanati, Kenig and Vanden Eynden [2], the same condition as the *gracious* labelling of Grannell, Griggs and Holroyd [4], and the β^+ -labeling of [5, A342225]. The question we answer is the first bullet of the ‘‘Further Work’’ list of [1]. It reads, in full:

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Any graph which allows a near α -valuation must be bipartite. Does every bipartite graph that possesses a β -valuation have a near α -valuation? We have examples of β -valuations on bipartite graphs which are not near α -valuations, but for each there exists a different vertex labelling which is a near α -valuation.

The third sentence is part of the question and fixes its level: what is asked is not whether a given β -valuation is a near α -valuation — the authors already have such examples — but whether *some* β -valuation of the graph is one. That is the statement refuted below.

The ambient convention. The standing convention of [1] is that “our graphs are finite and simple with no loops or isolated vertices”. Nothing is assumed connected — the word *connected* occurs there only inside the definition of a sun graph — and no bipartition is fixed in advance. Disconnected graphs are in scope: the illustrative non-example given immediately after the definition of a β -valuation is $P_1 \cup P_1$, and β -valuations of the disconnected graphs $2C_{4k}$ and $2C_{4k+2}$ are computed there. This matters, because every counterexample below is disconnected; see Section 4.

2. THE COUNTEREXAMPLE

Let

$$\begin{aligned} G &= C_4 \cup P_3, & V(G) &= \{v_1, v_2, \dots, v_7\}, \\ E(G) &= \{v_1v_2, v_2v_3, v_3v_4, v_4v_1\} \cup \{v_5v_6, v_6v_7\} : \end{aligned}$$

a 4-cycle together with a disjoint path on three vertices. Then $|V(G)| = 7$, $m = |E(G)| = 6 = |V(G)| - 1$, G is simple with no loop and no isolated vertex, and G is bipartite with parts $\{v_1, v_3, v_5, v_7\}$ and $\{v_2, v_4, v_6\}$. It is disconnected. Since $|V(G)| = m + 1$, every β -valuation of G is a bijection onto $\{0, 1, \dots, 6\}$.

Theorem 1. *$G = C_4 \cup P_3$ is a bipartite graph that possesses a β -valuation and has no near α -valuation. Consequently the answer to the question quoted above is negative.*

G is graceful. Put

$$\begin{aligned} b(v_1) &= 0, & b(v_2) &= 6, & b(v_3) &= 2, & b(v_4) &= 5, \\ b(v_5) &= 1, & b(v_6) &= 3, & b(v_7) &= 4, \end{aligned}$$

that is, the cycle $0 - 6 - 2 - 5 - 0$ and the path $1 - 3 - 4$. The six edge labels are 6, 4, 3, 5 and 2, 1, so they are exactly $\{1, 2, 3, 4, 5, 6\}$. (This half is not new: $C_m \cup P_n$ is graceful if and only if $m + n \geq 6$, recorded in Gallian’s survey [3, Table 1] with attribution to Traetta [6], and with P_n on n vertices our graph has $m + n = 7$. The labelling above makes the note self-contained.) Note that b itself is not a near α -valuation: v_6 carries 3 and its neighbours

carry 1 and 4. That is exactly the phenomenon [1] reports; the content of Theorem 1 is that no relabelling repairs it.

No β -valuation of G is a near α -valuation. We use one observation and then a table.

Lemma 2. *Let H have no isolated vertex and let b be an injective labelling of $V(H)$. Then a partition of $V(H)$ into V_{small} and V_{large} as in the definition exists if and only if every vertex is a local extremum, that is, carries a label strictly below all of its neighbours' or strictly above all of them.*

Proof. If the partition exists, each vertex of V_{small} is a local minimum and each vertex of V_{large} a local maximum. Conversely, let V_{small} be the set of local minima and V_{large} its complement, which consists of local maxima. The two sets are disjoint by construction, and a vertex cannot be both a local minimum and a local maximum because it has a neighbour. $\square$

Proof of Theorem 1. Let b be a β -valuation of G that is a near α -valuation, so by Lemma 2 every vertex is a local extremum. Recall that b is a bijection onto $\{0, \dots, 6\}$.

Step 1: the cycle. On C_4 the local minima form a set meeting every edge in exactly one endpoint, so they are one of the two bipartition classes $\{v_1, v_3\}$, $\{v_2, v_4\}$; the other class consists of local maxima. Hence the four labels on the cycle split as a low pair $\{p, q\}$ and a high pair $\{r, s\}$ with

$$p < q < r < s,$$

and, since $C_4 = K_{2,2}$, the four cycle edge labels are the four differences

$$D = \{s - p, r - p, s - q, r - q\}.$$

They must be pairwise distinct. Note for later that $s - p$ is the largest and $r - q$ the smallest of the four, and that $(s - p) + (r - q) = (r - p) + (s - q)$.

Step 2: the two labels 0 and 6 are adjacent. The difference 6 must occur as an edge label, and the only pair of labels in $\{0, \dots, 6\}$ at distance 6 is $\{0, 6\}$.

Step 3: fifteen cases. The vertex labelled 0 can only be a local minimum and the vertex labelled 6 only a local maximum. By Step 2 they lie in the same component.

If both lie on the cycle, then $p = 0$ and $s = 6$, so the cycle carries the labels $\{0, q, r, 6\}$ with $0 < q < r < 6$: ten possibilities. If both lie on the path, the cycle carries a 4-element subset L of $\{1, 2, 3, 4, 5\}$, and the only splitting of L into a low pair below a high pair is (two smallest | two largest): five possibilities.

In each of these fifteen cases the four cycle labels, hence D , are determined, and so are the three labels left for the path $x - y - z$ and the two edge labels $\{1, \dots, 6\} \setminus D$ that the path must realise. The path contributes the edge labels $|y - x|$ and $|y - z|$, and its centre y must be a local extremum, i.e. y must not lie strictly between x and z . Table 1 lists all fifteen cases; every one of them is impossible, for one of three reasons:

- R** two of the four cycle edge labels coincide, so b is not a β -valuation;
N no choice of centre among the three remaining labels realises the required pair of path edge labels;
X exactly one choice of centre realises it, and that centre lies strictly between its two neighbours, so it is not a local extremum.

Five cases fall to **R**, eight to **N** and two to **X**. No case survives, so no such b exists. $\square$

	cycle labels	cycle edge labels D	path labels	path needs	
A	$\{0, 1, 2, 6\}$	1, 2, 5, 6	$\{3, 4, 5\}$	$\{3, 4\}$	N
A	$\{0, 1, 3, 6\}$	2, 3, 5, 6	$\{2, 4, 5\}$	$\{1, 4\}$	N
A	$\{0, 1, 4, 6\}$	3, 4, 5, 6	$\{2, 3, 5\}$	$\{1, 2\}$	X
A	$\{0, 1, 5, 6\}$	4, 5, 5, 6	$\{2, 3, 4\}$	—	R
A	$\{0, 2, 3, 6\}$	1, 3, 4, 6	$\{1, 4, 5\}$	$\{2, 5\}$	N
A	$\{0, 2, 4, 6\}$	2, 4, 4, 6	$\{1, 3, 5\}$	—	R
A	$\{0, 2, 5, 6\}$	3, 4, 5, 6	$\{1, 3, 4\}$	$\{1, 2\}$	X
A	$\{0, 3, 4, 6\}$	1, 3, 4, 6	$\{1, 2, 5\}$	$\{2, 5\}$	N
A	$\{0, 3, 5, 6\}$	2, 3, 5, 6	$\{1, 2, 4\}$	$\{1, 4\}$	N
A	$\{0, 4, 5, 6\}$	1, 2, 5, 6	$\{1, 2, 3\}$	$\{3, 4\}$	N
B	$\{1, 2, 3, 4\}$	1, 2, 2, 3	$\{0, 5, 6\}$	—	R
B	$\{1, 2, 3, 5\}$	1, 2, 3, 4	$\{0, 4, 6\}$	$\{5, 6\}$	N
B	$\{1, 2, 4, 5\}$	2, 3, 3, 4	$\{0, 3, 6\}$	—	R
B	$\{1, 3, 4, 5\}$	1, 2, 3, 4	$\{0, 2, 6\}$	$\{5, 6\}$	N
B	$\{2, 3, 4, 5\}$	1, 2, 2, 3	$\{0, 1, 6\}$	—	R

TABLE 1. The fifteen cases in the proof of Theorem 1. Case A is “0 and 6 on the cycle”, case B is “0 and 6 on the path”. The two **X** rows are the only ones where the required path edge labels can be realised at all, and in both of them the forced centre is 3, lying strictly between 2 and 5, respectively between 1 and 4.

3. EDGE-MINIMALITY, AND THE CENSUS TO TEN EDGES

Because a graceful graph with m edges has at most $m + 1$ vertices, the class of graphs to be searched at each m is finite and can be reached by enumerating labellings rather than graphs: a graceful labelled edge set with m edges is exactly a choice, for each difference $d \in \{1, \dots, m\}$, of one of the $m - d + 1$ pairs $\{a, a + d\} \subseteq \{0, \dots, m\}$, so there are precisely $m!$ of them. Classifying these up to isomorphism and recording, for each class, whether any of its graceful labellings is a near α -valuation yields Table 2.

Theorem 3. *Up to isomorphism, $C_4 \cup P_3$ is the only graph with at most 6 edges that is bipartite, graceful and has no near α -valuation. In particular*

no counterexample to the question of [1] has fewer than 6 edges: each of the 15 bipartite graceful isomorphism classes with at most 5 edges has a near α -valuation, and 18 have exactly 6 edges, of which exactly one fails.

The qualifier “up to isomorphism” is needed: $C_4 \cup P_3$ alone accounts for 32 graceful labellings, in two orbits under its automorphism group of order 16.

Theorem 4. *Up to isomorphism, $C_4 \cup P_3$ ($m = 6$) and $C_6 \cup K_{1,4}$ ($m = 10$) are the only bipartite graceful graphs with at most 10 edges having no near α -valuation. Both are disconnected, and each has exactly $m + 1$ vertices.*

For $C_6 \cup K_{1,4}$, take the labelled cycle $0 - 9 - 5 - 3 - 2 - 10 - 0$ together with the star of centre 1 and leaves 4, 6, 7, 8. The cycle edge labels are 9, 4, 2, 1, 8, 10 and the star edge labels are 3, 5, 6, 7, together $\{1, \dots, 10\}$; so this graph too is graceful. That none of its 576 β -valuations is a near α -valuation is part of the census, and is not proved by hand here.

m	classes	bipart.	no near α	m	classes	bipart.	no near α
1	1	1	0	6	37	18	1
2	1	1	0	7	112	44	0
3	3	2	0	8	340	119	0
4	5	4	0	9	1078	323	0
5	12	7	0	10	3620	919	1

TABLE 2. Isomorphism classes of graceful graphs with m edges and no isolated vertex; how many are bipartite; and how many of those have no near α -valuation. The accompanying program re-derives every row; of the 1438 bipartite classes listed, 1052 are connected and 386 are disconnected.

4. WHAT IS NOT SETTLED

The connected question. Both counterexamples of Theorem 4 are disconnected, and the refutation therefore depends on the convention of [1] quoted in Section 1, under which graphs are not assumed connected. The strengthened question

does every *connected* bipartite graph possessing a β -valuation have a near α -valuation?

is untouched by this note. What the census does say is that the answer is yes for all 1052 connected bipartite graceful isomorphism classes with at most 10 edges. Gallian frames the equivalent notion of a *gracious* labelling explicitly for connected bipartite graphs, with the partite sets fixed in advance [3, §3.1]; under that reading of “bipartite graph” this note is a scope correction together with a small-case verification. The tree sub-case is a conjecture of

El-Zanati, Kenig and Vanden Eynden [2], restated for gracious labellings by Grannell, Griggs and Holroyd [4] and verified by them to 20 vertices; our census confirms it for every graceful tree with at most 10 edges.

Eleven edges and beyond are unswept. Nothing here bounds the number of counterexamples with $m \geq 11$.

Vertex-minimality is not claimed. Theorem 3 minimises the number of edges. Graphs on at most 6 vertices with more than 6 edges were not examined, so nothing here asserts that 7 is the least possible number of vertices.

5. VERIFICATION

The reader needs nothing but Section 2 to check Theorem 1: six differences by eye for the graceful labelling, and fifteen rows of Table 1. A program `verify.py` accompanies this note for the rest. It is Python 3.9+ using the standard library only, with no external data file and no floating-point arithmetic, and it reads the objects printed above. It re-derives: the structural claims about G ; the exhibited labelling; the equivalence of Lemma 2, over all 5040 labellings of G ; all fifteen rows of Table 1 including their reason codes; independently, by brute force over all 5040 bijections, that G has exactly 32 β -valuations of which none is near α and none is α ; all 576 β -valuations of $C_6 \cup K_{1,4}$ and the absence of a near α -valuation among them; every row of Table 2, together with the identification of the 6-edge and 10-edge counterexamples; and a list of named controls of both polarities.

What that program does *not* cover is stated in its own output: any $m \geq 11$; the connected question; and vertex-minimality.

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